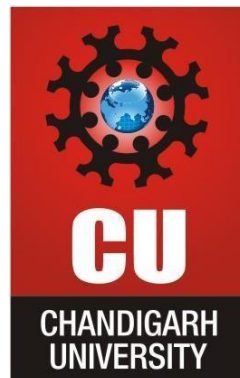




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Assignment - 2

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Subject Name: Computer Networks

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Section/Group: 24AIT_KRG-G2
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Q1. What is an IP Address? Why do we need classes in case of IPV4 address? Explain each class, Mention how many host and networks are possible in each class. Also explain what is subnetting and supernetting with examples.

Ans:

An IP (Internet Protocol) address is a unique logical address assigned to a device on a network. It helps identify the source and destination of data.

IPv4 is a 32-bit address, written in four octets.

Example: 192.168.1.10

Why are classes used in IPv4?

IPv4 classes were introduced to divide the address space according to the size of networks. Different organizations could therefore use address ranges suitable for small, medium, or large networks.

IPv4 Address Classes

Class	Range	Default Mask	Networks	Hosts/Network
A	1–126	255.0.0.0 (/8)	126	16,777,214
B	128–191	255.255.0.0 (/16)	16,384	65,534
C	192–223	255.255.255.0 (/24)	2,097,152	254
D	224–239	Multicast	—	—
E	240–255	Experimental	—	—

Class A: Large networks. Example: 10.0.0.1

Class B: Medium-sized networks. Example: 172.16.0.1

Class C: Small networks. Example: 192.168.1.1

Class D: Used for multicast.

Class E: Reserved for experimental purposes.

Subnetting : Subnetting is the process of dividing one large network into smaller networks called subnets.

Example:

192.168.1.0/24 → 192.168.1.0/26

A /26 gives:

- Number of subnets = $2^2 = 4$
- Usable hosts per subnet = $2^6 - 2 = 62$

Benefits: Better IP utilization, reduced broadcast traffic, and easier network management.

Supernetting: Supernetting combines multiple smaller networks into one larger network. It is also called route aggregation.

Example:

192.168.0.0/24

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

can be summarized as:

192.168.0.0/22

It reduces the size of routing tables.

Q2. What is the difference between Link state and Distance vector routing? Explain how these two algorithms help in identifying the least cost path with the help of examples.

Ans:

Distance Vector Routing In Distance Vector routing, a router shares information about the distance/cost to destinations with its neighbouring routers.

It commonly uses the Bellman-Ford algorithm.

Link State Routing : In Link State routing, routers build a map of the network topology and calculate the shortest path using Dijkstra's algorithm.

Possible paths:

- $A \rightarrow C = 8$
- $A \rightarrow B \rightarrow C = 2 + 3 = 5$

Therefore, the least-cost path is:

$A \rightarrow B \rightarrow C$, cost = 5

Difference

Feature	Distance Vector	Link State
Knowledge	Mainly neighbors	Network topology
Algorithm	Bellman-Ford	Dijkstra
Updates	To neighbors	Link-state information
Convergence	Generally slower	Generally faster
Example	RIP	OSPF

Q3. What is congestion in a network? Why do we need to control congestion? the difference between Leaky bucket and Token bucket with examples. Also, explain why do we need to do fragmentation?

Ans:

What is Congestion?

Network congestion occurs when the amount of traffic entering a network is greater than its capacity to process it.

For example, if a router can handle 100 Mbps but receives 150 Mbps, congestion may occur.

Why Control Congestion?

Congestion control helps to:

- Reduce packet loss
- Reduce delay
- Prevent buffer overflow
- Improve network performance
- Maintain efficient data transmission

Leaky Bucket

The Leaky Bucket controls traffic by sending packets at a relatively fixed rate, similar to water leaking from a bucket through a small hole.

Example:

If packets arrive in bursts but the output rate is 5 packets/sec, the algorithm sends them at approximately 5 packets/sec. If the bucket becomes full, excess packets may be discarded.

Token Bucket

The Token Bucket generates tokens at a fixed rate. A packet can be transmitted when a token is available.

It allows controlled bursts when tokens have accumulated.

Example:

If 10 tokens have accumulated, up to 10 packets can potentially be sent in a burst, subject to the configured limits.

Difference

Feature	Leaky Bucket	Token Bucket
Output	Mostly fixed rate	Variable rate
Bursts	Not normally allowed	Controlled bursts allowed
Mechanism	Queue/bucket	Tokens
Excess traffic	May be discarded	Can wait/use accumulated tokens

Fragmentation

Fragmentation is the process of dividing a large IP packet into smaller fragments so that it can pass through a network with a smaller MTU (Maximum Transmission Unit).

Example:

If a packet is 4000 bytes but the next network has an MTU of 1500 bytes, the packet may need to be divided into smaller fragments.

Need for fragmentation:

- Allows packets to cross networks with smaller MTUs.
- Supports communication between networks with different MTU sizes.
- Prevents oversized packets from being unable to cross a link.